

P2 — Neural Macro Compilation (Shortcut Compilation)

: v0.1 | : 2026-08-11 | P1 §6 , /

1. Introduction

- : / , (: steps/tokens/latency/KV)
- : (teacher→student) few-shot → (Learning to Compile Programs to Neural Networks, ICML 2024) — / / teacher
- : → , , fallback

2. Setting

- (): , ,
- ():
- : $\text{Sem_fast}() = \text{Sem_expanded}(T_)$, $\text{Cost_fast} \leq \text{Cost_expanded}$

3. Method (semantic-path self-distillation)

1. : $(E() = T_)$
2. : (/oracle) (, x_i , y_i),
3. : (embeddings/adapter/ / —)
4. : vs ()
5. fallback: ; execution-cost threshold

4. Results

- : , $\text{Cost} \leq \text{EXP-51}$
- : = (EXP-52)
- : / (EXP-50)

5. Ablations

- : 10/20/50
- : embeddings-only vs adapter vs ()
- : fallback

6. Related Work

- Program→NN compilation (ICML 2024) / inference compilation / shortcut distillation (diffusion): = + + fallback

7. Limitations

- ()
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- M1: (_G1 / AI .md)

(, 2026-08-11)

- , , :
- ($\rightarrow$: token ; attention);
- **CoT** **flashing** ($\rightarrow$: ; synthetic CoT);
- ($\rightarrow$: , ; distilled intuition / flashing).
- , — ASI ,